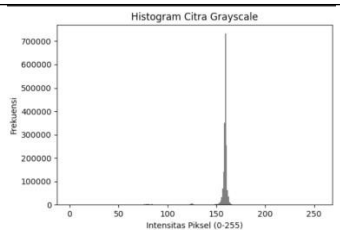
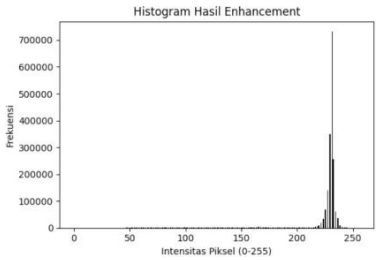
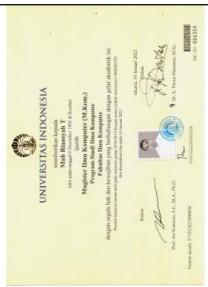
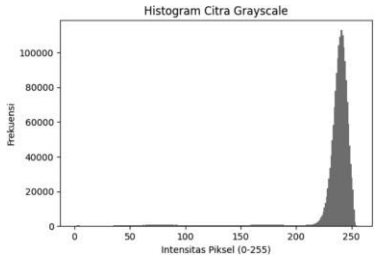
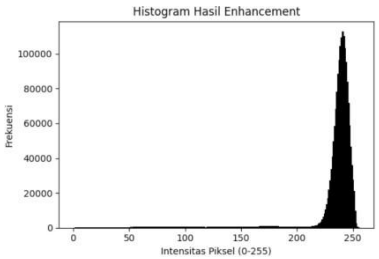
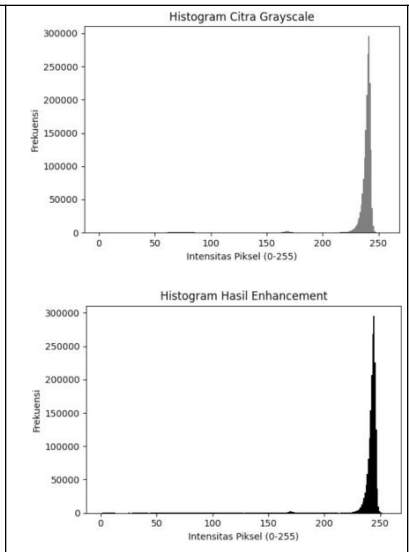
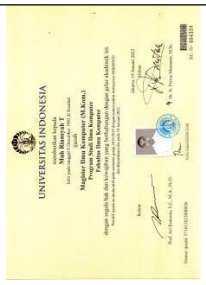
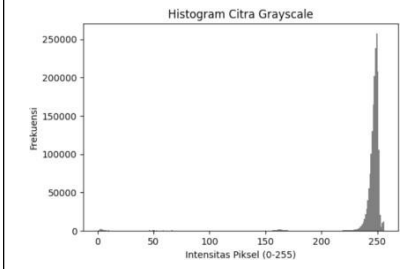
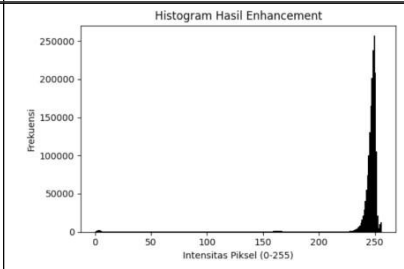
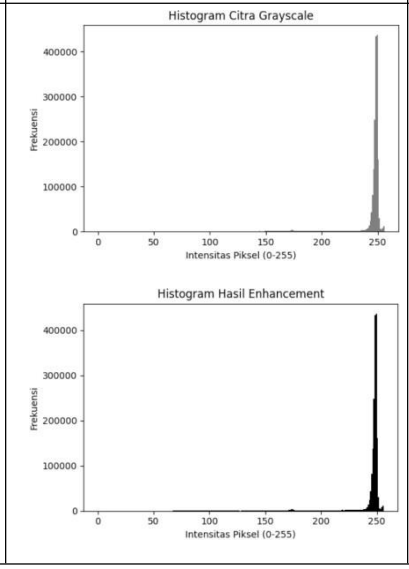
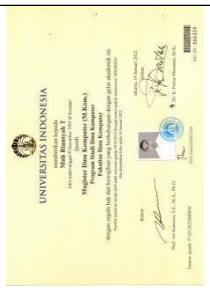
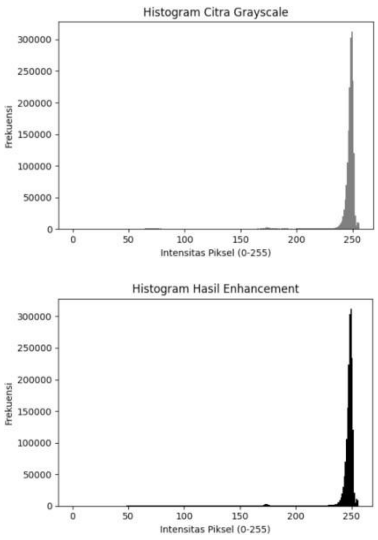
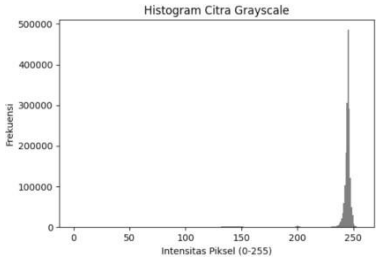
TUGAS 3 PENGOLAHAN CITRA DIGITAL

Analisis distribusi intensitas piksel

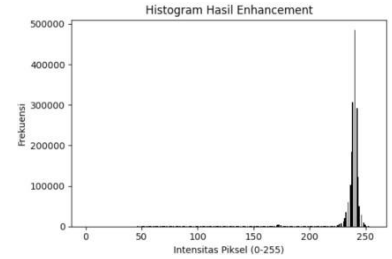
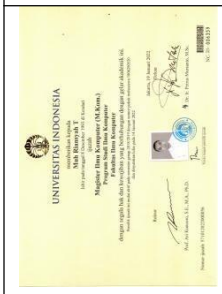
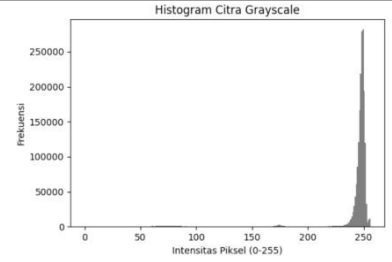
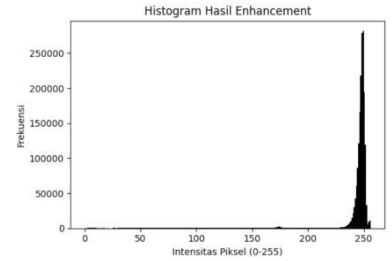
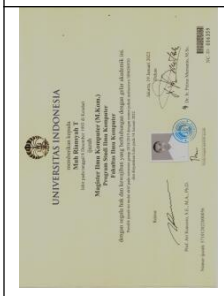
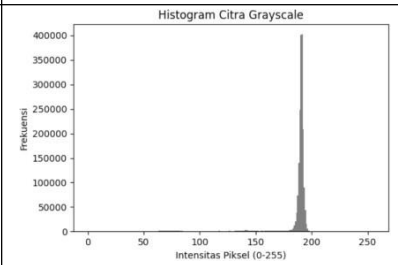
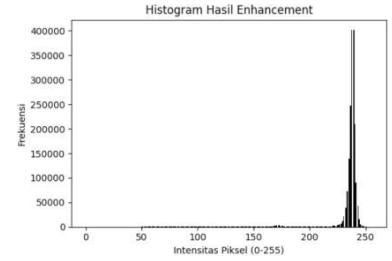
Nama : Fhaiz Rahmady Zaky
NIM : F1G124030
Kelas : C

Gambar Citra	Karakteristik Visual & Histogram	Identifikasi Masalah	Metode Enhancement	Alasan Pemilihan Metode
		Terlalu Gelap (Under-exposed) & Kontras Rendah	Gamma Correction (γ < 1.0) / Contrast Stretching	Menerangkan citra secara non-linear dan melebarkan rentang intensitas agar latar
				belakang terang tanpa mengaburkan teks.
	 	Kontras Rendah (Low Contrast)	Contrast Stretching (Peregangan Kontras)	Meregangkan rentang intensitas piksel agar pemisahan antara teks hitam dan latar belakang krem menjadi tegas.

		Pergeseran Warna (Color Cast) & Redup	Grayscale Conversion (Luminance) + Brightening	Konversi grayscale membuang distorsi warna, dilanjutkan peningkatan kecerahan agar latar menjadi bersih.
		Terlalu Terang (Over-exposed)	Gamma Correction ($\gamma > 1.0$) / Linear Intensity Slicing	Menggeser intensitas ke arah gelap untuk mempertegas tulisan yang tampak pudar akibat over-exposure.
				
		Terdistorsi Noise	Median Filter / Gaussian Blur	Median Filter sangat efektif menghilangkan salt-and-pepper noise tanpa mengaburkan garis tepi karakter teks.

		Kontras Kurang Optimal untuk OCR	Adaptive Thresholding (Otsu Thresholding)	Mengubah citra menjadi biner murni (hitam-putih) sehingga latar belakang menjadi putih bersih dan teks hitam pekat.
		Pudar / Kontras Rendah	Histogram Equalization (HE)	Meratakan distribusi frekuensi intensitas piksel secara global agar detail teks dokumen lebih menonjol.

Histogram yang lebih tersebar berpotensi meningkatkan kontras dan keterlihatan detail, tetapi bukan jaminan mutlak kualitas lebih baik. Kualitas citra yang baik bergantung pada keseimbangan antara kontras, natural appearance, preservasi informasi asli, dan tujuan akhir penggunaan citra tersebut.

				
	 	Pencahayaan Tidak Merata (Uneven Illumination)	CLAHE (Contrast Limited Adaptive Histogram Equalization)	Memperbaiki kontras secara lokal (block-by-block) sehingga bayangan/ketidak Merataan cahaya dapat diratakan.
	 	Kondisi Normal / Baik	Unsharp Masking (Penajaman Citra)	Meningkatkan ketajaman (sharpening) pada batas tepi karakter teks untuk memperjelas keterbacaan.